(ii) If $\rho\left(\left[A, B\right]\right)=\rho\left(A\right)=n$, then the equations are consistent and have unique solution.

(iii) If $\rho\left(\left[A, B\right]\right)=\rho\left(A\right)<n$, then the equations are consistent and have infinitely many solutions.

(iv) If $\rho\left(\left[A, B\right]\right)\neq\rho\left(A\right)$ then the equations are inconsistent and has no solution.

Example 1.9

Show that the equations $x+y=5$, $2x+y=8$ are consistent and solve them.

Solution:

The matrix equation corresponding to the given system is

$$\begin{pmatrix}1 & 1\\ 2 & 1\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}5\\ 8\end{pmatrix}$$

$$A \quad X \quad = \quad B$$

Matrix A	Augmented matrix $[A,B]$	Elementary Transformation
$\begin{pmatrix}1 & 1\\ 2 & 1\end{pmatrix}$	$\begin{pmatrix}1 & 1 & 5\\ 2 & 1 & 8\end{pmatrix}$	
$\sim\begin{pmatrix}1 & 1\\ 0 & -1\end{pmatrix}$	$\sim\begin{pmatrix}1 & 1 & 5\\ 0 & -1 & -2\end{pmatrix}$	$R_2 \to R_2 - 2R_1$
$\rho\left(A\right)=2$	$\rho\left(\left[A,B\right]\right)=2$	

Number of non-zero rows is 2.

$\rho\left(A\right)=\rho\left(\left[A,B\right]\right)=2=$ Number of unknowns.

The given system is consistent and has unique solution.

Now, the given system is transformed into

$$\begin{pmatrix}1 & 1\\ 0 & -1\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}5\\ -2\end{pmatrix}$$

$$x+y=5 \qquad (1)$$

$$y=2$$

$$\therefore (1) \Rightarrow x+2=5$$

$$x=3$$

Solution is $x=3,\ y=2$

Example 1.10

Show that the equations $2x+y=5$, $4x+2y=10$ are consistent and solve them.

Solution:

The matrix equation corresponding to the system is

$$\begin{pmatrix}2 & 1\\ 4 & 2\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}5\\ 10\end{pmatrix}$$

$$A \quad X \quad = \quad B$$

Matrix A	Augmented matrix $[A,B]$	Elementary Transformation
$\begin{pmatrix}2 & 1\\ 4 & 2\end{pmatrix}$	$\begin{pmatrix}2 & 1 & 5\\ 4 & 2 & 10\end{pmatrix}$	
$\sim\begin{pmatrix}2 & 1\\ 0 & 0\end{pmatrix}$	$\sim\begin{pmatrix}2 & 1 & 5\\ 0 & 0 & 0\end{pmatrix}$	$R_2 \to R_2 - 2R_1$
$\rho\left(A\right)=1$	$\rho\left(\left[A,B\right]\right)=1$	

$\rho\left(A\right)=\rho\left(\left[A,B\right]\right)=1<$ number of unknowns

$\therefore$ The given system is consistent and has infinitely many solutions.

Now, the given system is transformed into the matrix equation.

$$\begin{pmatrix}2 & 1\\ 0 & 0\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}5\\ 0\end{pmatrix}$$

$$\Rightarrow 2x+y=5$$

Let us take $y=k,\ k\in R$

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